regulatory reporting requirements.

- Conduct regular drills to ensure readiness for potential security incidents.

Current best practices for security

To stay ahead of evolving threats, industrial cloud environments adopt best practices and leverage emerging technologies.

Artificial intelligence and machine learning:

- Utilise AI/ML-powered security tools for anomaly detection and predictive threat analysis.
- Implement behavioural analytics to identify unusual patterns that may indicate security threats.

Containerisation and microservices security:

- Secure containerised applications and microservices architectures using specialised tools and practices.
- Implement runtime application self-protection (RASP) for containerised environments.

Cloud security posture management (CSPM):

- Employ CSPM tools to continuously monitor and manage cloud security risks.
- Automate security policy enforcement across multi-cloud environments.

Quantum-safe cryptography:

- Plan for the transition to quantum-safe cryptographic algorithms to protect against future quantum computing threats.

Automated compliance and security testing:

- Implement automated compliance checks and security testing as part of the CI/CD pipeline.
- Utilise Infrastructure as Code (IaC) security scanning tools to identify misconfigurations before deployment.

Edge computing security:

- Extend security measures to edge devices and gateways in industrial IoT environments.
- Implement secure boot and trusted execution environments for edge devices.

Blockchain for supply chain security:

- Explore blockchain technology for enhancing supply chain transparency and security in industrial settings.

The industry cloud from Microsoft

Microsoft's industry cloud is an excellent solution for industrial companies that want to leverage the power of cloud computing, artificial intelligence, and the Internet of Things to optimise their operations, enhance their productivity, and innovate faster. It offers the following benefits:

- Provides a seamless and consistent experience across different devices, platforms, and environments, such as Windows, Linux, iOS, Android, and Azure.
- Supports a wide range of industrial scenarios and use cases, such as predictive maintenance, quality control, asset management, process optimisation, worker safety, and customer service.
- Integrates with various Microsoft products and services, such as Microsoft 365, Dynamics 365, Power Platform, Azure IoT, Azure Synapse Analytics, Azure Machine Learning, and Azure Cognitive Services, as well as third-party solutions and partners, to deliver a comprehensive and end-to-end industrial solution.
- Empowers and enables industrial workers with the right tools and information to perform their tasks more efficiently and effectively, such as digital twins, mixed reality, chatbots, voice assistants, and intelligent recommendations.
- Ensures high levels of security, privacy, compliance, and reliability for industrial data and applications, using various features and capabilities, such as encryption, authentication, authorisation, auditing, backup, recovery, and resilience.
- Leverages the power of artificial intelligence, machine learning, and data analytics to optimise industrial processes, improve product quality, enhance customer satisfaction, and generate new business insights and opportunities.
- Supports the development and deployment of innovative and scalable industrial applications and solutions, using various services and tools, such as Azure IoT, Azure Synapse, Azure Machine Learning, and Azure DevOps.
- Fosters a culture of innovation and collaboration among industrial organisations and their ecosystem partners using various platforms and communities, such as GitHub and Microsoft Learn.

This future-ready solution can help industrial companies achieve their digital transformation goals and stay ahead of the competition. END

References

- *https://industrial-cloud.com/*
- *https://www.avanade.com/en/technologies/microsoft-industry-clouds*
- *https://www.forrester.com/what-it-means/ep375-industry-clouds/*

By: Dr Magesh Kasthuri and Dr Anand Nayyar

Dr Magesh Kasthuri is a senior distinguished member of the technical staff and principal consultant at Wipro Ltd. This article expresses his views and not that of Wipro.

Dr Anand Nayyar is a PhD in wireless sensor networks and swarm intelligence. He works at Duy Tan University, Vietnam, and loves to explore open source technologies, IoT, cloud computing, deep learning, and cyber security.